

# Infoblox DNS Cache Acceleration Administrator Guide

For NIOS 9.0.X
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# Contents

**About DNS Cache Acceleration ..... 4**
Supported DNS Cache Acceleration Appliances ..... 4
Operational Guidelines ..... 4
DNS Cache Acceleration Appliance System Configuration ..... 7
Licensing Requirements ..... 7
Configuring DNS Cache Acceleration ..... 8
Enabling the DNS Cache Acceleration Service ..... 8
Enabling the DNS Service ..... 9
Specifying Grid-Level DNS Cache Acceleration Settings ..... 9
Specifying Member-Level DNS Cache Acceleration Settings ..... 9
**Supported Features for DNS Cache Acceleration Appliances ..... 11**
Managing Query Performance ..... 11
Preserving the RRset Order for Cached DNS Responses ..... 11
Defining Sort List for Cached DNS Responses ..... 12
Clearing DNS Cache ..... 13
Clearing DNS Cache ..... 13
Clearing Cache for DNS Views ..... 13
Clearing Domains and Subdomains from DNS Cache ..... 14
Viewing the DNS Accelerator Cache ..... 14
Configuring Forward and Stub Zones ..... 15
Infoblox DNS Firewall ..... 15
VLAN Tagging support ..... 15
Limitations of VLAN on DCA Capable Appliances ..... 15
Enabling VLAN Support Using CLI Commands ..... 16
DSCP support ..... 18
Configuring DSCP Value using CLI Commands ..... 19
**Monitoring Cache Acceleration Statistics ..... 21**
DNS Statistics for Cache Acceleration ..... 21
Reports for DNS Cache Acceleration Appliances ..... 21
Supported Reports for DNS Cache Acceleration Appliances ..... 22
DNS Cache Acceleration CLI Commands ..... 22
set temp_license ..... 23
show license ..... 23

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set dns ..... 23
show dns ..... 24
set monitor ..... 25
show monitor dns ..... 25
set dns-accel log ..... 25
set dns-accel log ..... 26
show dns-accel ..... 26
show dns-accel ..... 27
show dns-accel-cache ..... 27
show dns-accel-cache ..... 28
show dns-accel-cache [existing-file] ..... 29
show dns-accel-cache [existing-file] ..... 29
set dns-accel-debug ..... 30
show dns-accel-debug ..... 30
set vdca_type64_support ..... 30
set vdca_type65_support ..... 31
show vdca_type64_support ..... 32
show vdca_type65_support ..... 32

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# About DNS Cache Acceleration

Using the DNS Cache Acceleration service (DCA), can make DNS query resolution more responsive for resolvers across the network. These appliances can be part of an Infoblox Grid or operated as independent DNS-caching appliances. The appliances support DNS queries on the LAN1, LAN2, MGMT, and HA ports.

## Supported DNS Cache Acceleration Appliances

Following hardware appliances support software-based DNS Cache Acceleration:

- Trinzic X5 Series
    - IB-22x5 series
    - IB-40x5 series
- Trinzic X6 Series
    - TE-2306 series
    - TE-4106 series

Following virtual appliances support software-based DNS Cache Acceleration:

- Trinzic X5 Series
    - IB-v22x5
    - IB-v40x5
- IB-FLEX
- Trinzic X6 Series
    - TE-2326
    - TE-4126

## Operational Guidelines

The specialized function of appliances that support DNS Cache Acceleration is to act as a high-speed DNS caching-only name server. The DNS Cache Acceleration appliances share the following characteristics:

- The DNS Cache Acceleration appliances support the following:
    - Up to six DNS views
    - Forward zones and stub zones, but not authoritative zones
    - Certain Finisar Copper and Fiber SFP modules
    - Anycast for BGP v6 and OSPF v3
    - DNS Anycast and IPv6 Anycast
    - Up to 10,000 entries for each ACL (Access Control List)
    - Only the cyclic ordering for A records over the IPv4 transport
- The LAN1, LAN2, MGMT and HA interfaces of DNS Cache Acceleration appliances support IPv4 and IPv6 transports and DNS services over IPv4 and IPv6 networks.
- The DNS Cache Acceleration appliances support the following IPv6 functions and applications:
    - DNS over IPv6 LAN1, LAN2, MGMT, and HA interfaces
    - IP6 addresses on a loopback interface
    - CLI (SSH) access over IPv6
    - GUI access over IPv6
    - PAPI access over IPv6
    - Sending SNMP traps over IPv6
    - SNMP query over IPv6
    - Sending messages to an external syslog server over IPv6
    - Email relay over IPv6
    - IPv6 static routes
- When a NIOS appliance or the host restarts, you might continue to receive responses for cached queries from the DNS cache accelerator. Queries that are not cached will not be answered.
- If query logging is enabled, only DNS queries will be logged.
- The DNS Cache Acceleration appliances do not support the following:

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- DHCP and IPAM functions
- Zone transfers or dynamic DNS updates

In cache-accelerated mode, the DNS Cache Acceleration appliances have the following characteristics :

- They support DNS queries over IPv4 and IPv6 transports only for the following record types: A, AAAA, MX, PTR, and CNAME.
- When the DNS service restarts due to changes in the DNS configuration, all DNS caches are cleared.
- Appliances with cached acceleration do not support monitoring of DNS packets. They support DNS monitoring alerts and IP rate limiting.

The table below lists the features that are supported or not supported for the DNS Cache Acceleration feature on an appliance that supports DNS Cache Acceleration:

*Features on the DNS Cache Acceleration platforms*

| Features | Supported / Not Supported - Software-Based appliances |
|---|---|
| Tiered licensing | Supported<br>Note that only IB-4015 and TE-4126 support tiered licensing. |
| RPZ | Supported<br>For IB-FLEX appliances, only when you configure RPZ zones with DCA-enabled-flex-member, maximum cache lifetime is set to 300 seconds. |
| Caching (A, AAAA, MX, CNAME, PTR) | Supported |
| Caching of Type 64 and 65 resource records | Supported from NIOS 9.0.2 onwards |
| Resource records of type 64 and 65 (HTTPS and SVCB) | Supported from NIOS 9.0.2 onwards |
| Do not cache: EDNS, TCP, Any, TSIG | Supported |
| Caching over additional interfaces (v4, v6) | Supported |
| Dump Acceleration Cache (CLI, GUI, PAPI) | Supported |
| Clear Acceleration Cache (CLI, GUI, PAPI) | Supported |
| Cache pre-fetch and cache refresh | Supported |
| ACLs (Allow-queries/Responses, Match-Clients/Destination, Blackhole) | Supported |
| AAAA Filtering (Bypassed but support configuring) | Supported |
| Fixed RRSET ordering | Supported |
| DNS64 | Supported |
| DNS monitoring feature (netmon) | Supported |

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| Features | Supported / Not Supported - Software-Based appliances |
|---|---|
| DNS Query logging (BIND only) | Supported |
| DNS Views | Supported |
| Forward/Stub zones | Supported |
| DNS cache acceleration related restrictions for configuration. | Supported |
| Reporting | Supported, see *Supported Reports for DNS Cache Acceleration Appliances*. |
| VLAN | Supported |
| DSCP | Supported<br>DSCP is not supported when packets are processed by the DNS Cache Acceleration feature in software-based DNS cache acceleration appliances: IB-22*x*5, IB-V22x5, IB-40*x*5, IB-V40x5, TE-2326, TE-4126, IB-V2326, IB-V4126 |
| Sort list | Supported |
| Anycast (OSPF and BGP) | Supported |
| BFD (Bidirectional Forwarding Detection) | Supported |
| HA Support | Supported |
| NIC Bonding | Supported |
| Multiple-Interfaces on same subnet | Not supported |
| IP Rate-limit and Response logging | Not supported |
| EDNS Client Subnet support | Not supported |
| NXDOMAIN redirection | Supported |
| DNSSEC (Bypassed but support configuring) | Supported |
| Debug enhancements | Supported |
| SNMP Support for DCA service related traps | Supported |
| SNMP stats support for DNS QPS and CHR | Supported |

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| Features | Supported / Not Supported - Software-Based appliances |
|---|---|
| NX Mitigation | Not supported |
| NetFilter (Tracking tables) | Supported |
| Traffic-capture (All modes) | Supported |
| No flush-mode support for DNS cache acceleration cache | Supported |
| Per-interface UDP DNS cache acceleration response counters | Supported |
| CLI commands | You can use the commands `set dns-accel` and `show dns-accel` to view and set the DNS Cache Acceleration. For more information, see *DNS Cache Acceleration CLI Commands*. |
| DNS Query rewrite (Bypassed but supports configuring) | Not supported |
| Threat Protection | Supported |

# DNS Cache Acceleration Appliance System Configuration

For initial hardware setup instructions, refer to the NIOS Documentation at *docs.infoblox.com*. Once you install the system and run the `set network` command, enable the DNS service and the DNS Cache Acceleration service on the DNS Cache Acceleration appliances.
The procedure to start DNS Cache Acceleration service consists of the following:

1. Install and activate the DNS Cache Acceleration license, as described in the *Licensing Requirements* section.
2. Enable the DNS service and the DNS Cache Acceleration service, as described in the Enabling the *DNS Cache Acceleration Service* and *Enabling the DNS Service* sections, see *Configuring DNS Cache Acceleration*.

## Licensing Requirements

The DNS Cache Acceleration appliance requires a **DNS Cache Acceleration** license. The IB-4015 and TE-4126 appliances support four tiers of **DNS Cache Acceleration** licenses. These licenses enable DNS caching and control DNS queries per second performance on the appliance. The Tier-4, Tier-3, and Tier-2 licenses enable the appliance to process up to 150K, 300K, and 600K qps (queries per second), respectively. The Tier-1 license enables the appliance to perform at its full capacity.

- If the appliance is running on NIOS 9.0, you can install either a temporary, subscription, or a permanent license on the appliance. For more information, see *Managing Licenses in NIOS 9.0.0* in the *Infoblox NIOS Documentation*.
- If the appliance is running on NIOS 9.0.1 or later, the appliance comes preinstalled with a temporary license and you can upgrade it to an appropriate subscription license. For more information, see *Managing Licenses in NIOS 9.0.1* in the *Infoblox NIOS Documentation*.

The temporary license provides a 60 days free trial for up to one million queries per second. You can upgrade a temporary license to a permanent or a subscription license depending on the NIOS version by using either Grid Manager or the CLI.

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> **Note**
> A lower tier license cannot be added for DNS Cache Acceleration when a higher tier permanent license is installed. You can add a lower tier license if a temporary license exists. You cannot downgrade any permanent DNS Cache Acceleration license from a higher tier license to a lower tier license. For more information about the different tier licenses, contact your Infoblox representative.
> The appliance displays a warning message if the DNS Cache Acceleration utilization exceeds a predefined threshold. You can configure the appliance for various levels of performance based on your current license type.
> Tiered licenses are supported only on IB-4105 appliances. Other software-based appliances are enabled to perform at their full capacity.

The system restarts automatically when you install the license or upgrade it to a higher tier. For more information about adding and upgrading licenses through Grid Manager, see the *Infoblox NIOS Documentation*.

You can also use the CLI command `set license` or the `set temp_license` command to add a license. For more information about the CLI commands, see *DNS Cache Acceleration CLI Commands*.

# Configuring DNS Cache Acceleration

To achieve optimal DNS query resolution performance, ensure that the switching infrastructure surrounding the DNS Cache Acceleration appliances can handle high performance gigabit ethernet traffic load. To operate efficiently, appliances require switches with adequate memory buffer capacity and support for 1000Base-TX full duplex operation.

> **Note**
> Operation of the DNS caching features requires a DNS Cache Acceleration license. For information about licensing requirements, *DNS Cache Acceleration Appliance System Configuration*.

The DNS service and the DNS Cache Acceleration service are disabled by default on NIOS appliances. They must be enabled after adding the appliance to the network. For information, see *Enabling the DNS Cache Acceleration Service* and *Enabling the DNS Service* sections below.

## Enabling the DNS Cache Acceleration Service

> **Note**
> If you start the DNS Cache Acceleration Service first. the appliance will also start the DNS service and display a notification that it is doing so.

To enable the DNS Cache Acceleration service on a DNS Cache Acceleration appliance, do the following:

1. From the **Grid** tab, select the **Grid Manager** tab and click **DNS Cache Acceleration**.
2. Select the *Grid_member* check box for the appliance.
3. Click the Start icon to activate the DNS Cache Acceleration service on the appliance.
   If you need to disable DNS Cache Acceleration service: click the Stop icon.

If the DNS Cache Acceleration appliance is an independent appliance, do the following:

1. From the **System** tab, select the **System Manager** tab and click **DNS Cache Acceleration**.
2. Click the Start icon to activate the DNS cache acceleration service on the appliance.

> **Note**
> The appliance displays a warning message if the average DNS Cache Acceleration utilization rate exceeds 85%.

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## Enabling the DNS Service

> **Note**
> The DCA (DNS Cache Acceleration) service is dependent upon the DNS service. Thus, when you start the DCA service, the DNS service will start automatically. However, if you start the DNS service first, you must start the DCA service separately. When both services are running on an appliance and you want to stop both services, you can select to stop the DCA service first and then stop the DNS service. If you stop only the DCA service, the DNS service will continue to run. In the same situation when both services are running and you stop the DNS service first, then the DCA service will stop automatically.

To enable the DNS name resolution service for an installed DNS Cache Acceleration appliance in an Infoblox Grid:

1. From the **Grid** tab, select the **Grid Manager** tab, ensure that DNS is selected and select the *Grid_member* check box.
2. Click the Start icon to activate the DNS service on the appliance.
   If you need to disable the DNS service, click the Stop icon.
3. Save the configuration.
4. Click **Restart** when it appears at the top of the screen.

If the appliance is an independent appliance for the network, do the following:

1. From the **System** tab, select the **System Manager** tab, ensure that DNS is selected and select the *Grid_member* check box.
2. Click the Start icon to activate the DNS service on the appliance. To disable DNS, click the Stop icon.
3. Save the configuration.
4. Click **Restart** if it appears at the top of the screen.

## Specifying Grid-Level DNS Cache Acceleration Settings

Each DNS response cached in the DNS Cache Acceleration appliance has its own TTL (Time-To-Live) value. For example, entering a value of 3599 seconds (1 hour-1 second) ensures that the appliance caches all DNS responses with a defined TTL of one hour (3600 seconds) or greater. All DNS responses with TTL values of 3599 seconds or less will not be cached by the appliance, and will instead be cached by non-accelerated cache server.
To specify Grid DNS Cache Acceleration settings:

1. From the **Data Management** tab, select the **DNS** tab, expand the Toolbar and click **Grid DNS Properties.**
2. In the *Grid DNS Properties* editor, click **DNS Cache Acceleration**.
3. By default, the **Don't Cache Records with TTL Less Than...** setting defaults to a value of One (1) second. The maximum permissible value is 65000 seconds (slightly over 18 hours).
4. Enter the new TTL value in the **Don't Cache Records with TTL Less Than...** field.
5. Save the configuration and click **Restart** to restart services.

## Specifying Member-Level DNS Cache Acceleration Settings

The TTL setting on the Member page applies only to an independent appliance.
To modify the TTL cache setting for the appliance, select the **Override** option. By doing so, the new caching value supersedes the value specified at the Grid level. If you select the **Override** option and do nothing else, the appliance inherits its settings from the Grid.
To specify DNS Cache Acceleration settings for the member:

1. From the **Data Management** tab, select the **DNS** tab -> **Members** tab -> *member* check box.
2. Click **Edit**.
3. In the *Member DNS Properties* editor, click **DNS Cache Acceleration**.
4. By default, the **Don't Cache Records with TTL Less Than...** setting is defined to inherit its value from the global Grid setting. The **Override** button is associated with this field.
5. To modify the TTL value, click **Override**. The field is enabled and the button changes to **Inherit**.

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6. Enter the new TTL value in the **Don't Cache Records with TTL Less Than...** field. The maximum permissible value for this setting is 65000 seconds (slightly over 18 hours).
7. To use the caching TTL value used across the Grid, click **Inherit**. The button changes to **Override** and the field is disabled.
8. Save the configuration.

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# Supported Features for DNS Cache Acceleration Appliances

The DNS Cache Acceleration appliances support the following features:

- *Managing Query Performance*
- *Preserving the RRset Order for Cached DNS Responses*
- *Defining Sort List for Cached DNS Responses*
- *Clearing DNS Cache*
- *Configuring Forward and Stub Zones*
- *Infoblox DNS Firewall*
- *DSCP support*

## Managing Query Performance

The appliance that supports DNS Cache Acceleration is configured by default to serve up to 1,000 concurrent clients that serve recursive queries. To improve query performance for the appliance, Infoblox recommends that you increase this setting to 40,000 concurrent clients.

1. From the **Data Management** tab, select the **DNS** tab and click the **Members** tab -> *member* check box -> Edit icon.
2. In the *Member DNS Properties* editor, click **Toggle Advanced Mode**.
3. When the additional tabs appear, click the **Advanced** subtab of the **Queries** tab.
4. Select the **Limit number of recursive clients to** option and enter the value **40000**.
5. Save the configuration.
6. Click **Restart** if it appears at the top of the screen.

## Preserving the RRset Order for Cached DNS Responses

By default, when a client queries a domain name, the DNS caching appliance returns the A and AAAA records of the domain name in a cyclic order. However, this default behavior can be overridden if you have enabled and configured (at the Grid level) fixed RRset order for hosts that have multiple addresses. When you override the default behavior and preserve the fixed RRset order for cached DNS responses, the DNS caching appliance returns A and AAAA records associated with domain names in the order they were received from an upstream server. You can specify the fixed RRset order for A, AAAA, or both A and AAAA records at the Grid level and override at the member and DNS view levels.
Note that configuring fixed RRset order for specific FQDNs might slightly affect the performance of the DNS caching appliance.
To preserve the fixed RRset order for cached DNS responses at the Grid, member, or DNS view level:

1. **Grid**: From the **Data Management** tab -> **DNS** tab, expand the Toolbar, and then click **Grid DNS Properties.**
   **Member**: From the **Data Management** tab, select the **DNS** tab and click the **Members** tab -> *member* check box -> Edit icon.
   **DNS View**: From the **Data Management** tab, select the **DNS** tab -> **Zones** tab -> *dns_view* check box -> Edit icon.
2. In the editor, select the **RRset Order** tab -> click the **Basic** tab, and then complete the following:
   - **Enable fixed RRset order for following FQDNs:** Select this check box to preserve the configuration of RRset order for cached DNS responses.
   - In the FQDN table, specify the list of FQDN entries for which you want to preserve the RRset order. Note that you can configure a maximum of 25 FQDNs for the specified RRset order.
     You can click the Add icon and complete the following to add a new entry to the list:
     - **FQDN:** Enter the fully qualified domain name with which the A or AAAA record is associated.
       **Note:** If you enter a wildcard character as part of the domain name, the appliance considers the wildcard character as a literal character. For example, if you enter test*.com, the appliance matches the domain name with test*.com only.
     - **Record Type:** Select the record type from the drop-down list. You can select **A**, **AAAA**, or **Both A and AAAA**.
3. Save the configuration and click **Restart** if it appears at the top of the screen.

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# Defining Sort List for Cached DNS Responses

You can define a sort list for the cached DNS responses on the appliance to prioritize A and AAAA records on certain networks, sorting them to the beginning of the list in the response. For example, you can define a sort list when a recursive server has two interfaces and you want the DNS clients to prefer one interface because it has a faster link. You can define sort lists for the cached DNS responses on the appliance, regardless of whether DNS cache acceleration license is installed or not.
Consider the following when the DNS cache acceleration license is installed on the appliance:

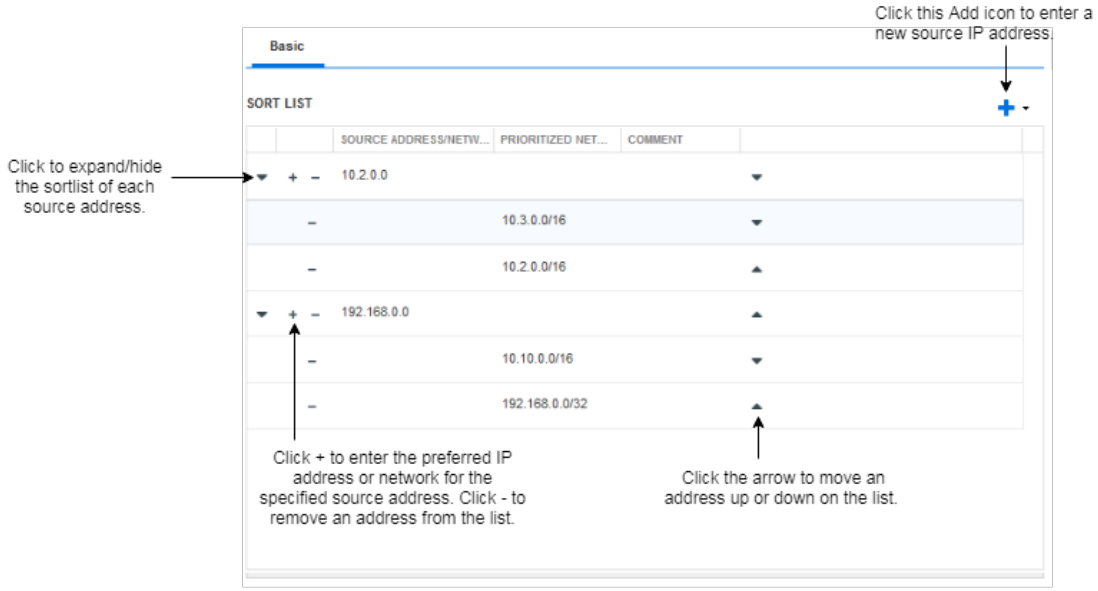
- Authoritative and stub zones are not supported by the appliance. Hence, sort lists are supported only for recursive caching.
- You can define up to 200 sort lists with a maximum of 50 IP addresses per sort list.
- A DNS view cannot be associated with the appliance if the DNS view has more than 200 sort lists or more than 50 IP addresses per sort list configured.

> **Note**
> If more than 200 sort lists or more than 50 IP addresses per sort list are configured for the cached DNS responses, you cannot upgrade the appliance to a later release.

When both RRset order and sort list are defined for cached DNS responses, the appliance sorts the responses based on the RRset order first, and then sorts based on the sort list. For information about preserving the RRset order for cached DNS responses, see *Preserving the RRset Order for Cached DNS Responses*. Note that defining sort lists for the cached DNS responses might have a performance impact on DNS caching.
To define a sort list for a Grid, member, or DNS view:

1. **Grid:** From the **Data Management** tab, select the **DNS** tab, expand the Toolbar and click **Grid DNS Properties.**
   **Member:** From the **Data Management** tab, select the **DNS** tab > **Members** tab -> *member* check box -> Edit icon.
   **DNS View:** From the **Data Management** tab, click the **DNS** tab -> **Zones** tab> *dns_view* check box -> Edit icon.
   To override an inherited property, click **Override** next to it and complete the appropriate fields.
2. In the editor, click **Toggle Advanced Mode**, and then click **Sort List**.
3. Click the Add icon and select either **Any** to define a sort list for any address and network, or **Address/Network** to define a sort list for a particular source IP address or network.
4. Do the following in the new row:
   - If you selected **Address/Network**, enter the IP address or network of the source of the query. The feature supports IPv4 or IPv6 values.
   - Click the Add icon beside the source IP address to add the preferred IP addresses or networks for the source. You can add as many IP addresses as necessary. However, if the DNS Cache Acceleration license is installed on the appliance, you can add a maximum of 50 IP addresses per sort list. When you add multiple IP addresses, you can change the order of the IP addresses. Select an IP address and drag it to its new position, or click the up or down arrow, as show in the following figure:
     *Sort List*

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5. Enter the IP address or network in the **Prioritized Networks** field.
6. You can add additional information about the source address or network in the **Comment** field.
7. To add another source IP address or network, click the Add icon again. You can create a separate sort list for each source IP address or network.
8. Save the configuration and click **Restart** if it appears at the top of the screen.

# Clearing DNS Cache

The DNS Cache Acceleration appliances allow you to clear certain information from the DNS cache. You can do the following:

- *Clearing DNS Cache*
- *Clearing Cache for DNS Views*
- *Clearing Domains and Subdomains from DNS Cache*
- *Viewing the DNS Accelerator Cache*

The appliance saves audit log entries for all the three clear cache operations. For information about audit logs, refer to the Infoblox NIOS documentation at *https://docs.infoblox.com*.

## Clearing DNS Cache

On the DNS Cache Acceleration appliance, you can clear all entries from the accelerator cache and the name server recursive cache.
To clear DNS cache:

1. From the **Data Management** tab, select the **DNS** tab -> **Members** tab -> *member* check box.
2. Expand the Toolbar, click **Clear** -> **Clear DNS Cache**.
3. Click **Yes** in the confirmation dialog box to clear DNS cache.

## Clearing Cache for DNS Views

This feature clears cache entries of a specific DNS view that is associated with the selected member.
To clear cache of a DNS view:

1. From the **Data Management** tab, select the **DNS** tab -> click the **Members** tab.

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2. Expand the Toolbar, click **Clear** -> **Clear View's Cache**.
3. Specify the following in the *Clear View's Cache* dialog box:
    - **Member**: Click **Select Member** to select a member. If there are multiple members, the *Member Selector* dialog box is displayed, from which you can select a member. Click the required member name in the dialog box. You can also click **Clear** to clear the displayed member and select a new one.
    - **DNS View**: Select a DNS view from the drop-down list. This list box appears only when there are multiple DNS views in the network view.
    - Click **Clear Cache** to clear the cache entries of the corresponding DNS View.

## Clearing Domains and Subdomains from DNS Cache

You can use this feature to clear specific domain and its subdomains from the DNS cache.
To clear a domain name:

1. From the **Data Management** tab, select the **DNS** tab -> click the **Members** tab.
2. Expand the Toolbar, click **Clear** -> **Clear Domain Name**.
3. Specify the following in the *Clear Domain Name from Cache* dialog box:
    - **Domain Name**: Enter a domain name you want to delete.
    - **Clear entire domain (including subdomains):** Select the check box to clear the specified domain and its subdomains from the DNS cache. For example, if you enter corp100.com in the **Domain Name** field, then selecting this check box clears the entire domain including its subdomains such as .xyz Domain Names | Join Generation XYZ, .xyz Domain Names | Join Generation XYZ, x.xyz.com, etc. This check box is deselected by default.
    - **Member**: Click **Select Member** to select a member. If there are multiple members, the *Member Selector* dialog box is displayed, from which you can select a member. Click the required member name in the dialog box. You can also click **Clear** to clear the displayed member and select a new one.
    - **DNS View**: Select a DNS view from the drop-down list. This list box appears only when there are multiple DNS views in the network view.
    - Click **Clear Domain Name** to clear the domain name from the cache.

## Viewing the DNS Accelerator Cache

You can view the cached contents of the DNS accelerator. The NIOS appliance temporarily disables the cache to obtain the current cache data. During this time, the non-accelerated name server on the appliance responds to all incoming DNS queries. Depending on the size of the cached data, it may take 40-50 seconds to compile information before the cache is re-activated.
Your browser may need to be set to allow popup windows from the IP address of the appliance.

The DNS Cache Acceleration processes incoming EDNS0 packets that contains the local ID, matches it with the DNS server as part of the RADIUS accounting message and populates the subscriber cache in DNS Cache Acceleration with the parental control policy information. For more information, refer to the NIOS documentation at *docs.infoblox.com.*

The cache data appears in plain text format, showing the following DNS response information:

- DNS View Number (0-5)
- Query question,Query type, and Query class
- Original TTL in seconds
- Remaining TTL in Seconds
- Cache Entry Usage Count
- Number of Seconds since the cache entry was last addressed
- Recursion
- Resource Record section, Type, Name, TTL, and Data
- Rcode indicates Response Code and displays 0 if there are no errors.
- Debug<1> and Debug<2> codes are used for debug purpose.
- PCP word - valid for subscriber services.
- Data based on the type of resource record(s) attached to the response:
    - A type and AAAA Filtered – IP address
    - CNAME, SOA, PTR – Domain Name

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- MX – Preference, Domain Name

To view the contents of the DNS Accelerator's cache, do the following:

1. From the **Data Management** tab, select the **DNS** tab and click the **Members** tab -> *member* check box.
2. Choose **View** -> **View Acceleration Cache**.
3. Click **Yes** in the *View Acceleration Cache* dialog.
   The system displays a *File Download was Successful* message and the cache data is displayed in table format in a new browser tab or browser window.

# Configuring Forward and Stub Zones

You can configure forward and stub zones on DNS Cache Acceleration appliances. A forward zone is where queries are sent before being forwarded to other remote name servers. A stub zone contains records that identify the authoritative name servers in another zone. For information about configuring forward and stub zones, see the NIOS Documentation at *docs.infoblox.com*.

# Infoblox DNS Firewall

Infoblox DNS Firewall employs DNS RPZ (Response Policy Zones), a technology developed by ISC (Internet System Consortium) for allowing reputable sources to dynamically communicate reputation domain names so you can implement policy controls for DNS lookups.
You can configure RPZs on a DCA capable member and define RPZ rulesets to block DNS resolutions for malicious or unauthorized domain names. For information
To configure Response Policy Zones, you must install a valid RPZ Enablement license.

To configure local RPZ and RPZ feed, see the NIOS Documentation at *docs.infoblox.com*. You can assign a DCA capable member that has DNS cache acceleration enabled as a Grid primary or secondary for the local RPZ. For an RPZ feed, you can assign a DCA capable member as a Grid secondary or Grid lead secondary.
The Local RPZs or RPZ feeds do not support DDNS updates. For more information, see the NIOS Documentation at *docs.infoblox.com.*

## VLAN Tagging support

The DNS Cache Acceleration appliance supports VLAN (Virtual Local Area Network) interfaces on LAN1, LAN2, and HA ports. Only the DNS service can listen on the VLAN interfaces. You can define default route on VLAN, LAN1 and LAN2 interfaces. For more information about configuring VLAN interface, see the NIOS Documentation at *docs.infoblox.com.*
You can use the `set network` command to set the primary IPv4 and IPv6 LAN1 networks as a tagged or an untagged VLAN interface. To view the VLANs on each port, use the `show interface` command. The appliance allows you to use the VLAN interface from LAN1 to be used as an advertising interface for OSPF. For more information, see the NIOS Documentation at *docs.infoblox.com.*

## Limitations of VLAN on DCA Capable Appliances

Following are the limitations of VLAN interface on a DNS Cache Acceleration appliance:

- The appliance does not support overlap of addresses or networks across VLANs.
- You cannot update member network properties when a scheduled upgrade is in progress.
- The appliance does not support SNMP interface statistics and traffic report statistics for accelerated DNS traffic over tagged VLANs as this may lead to performance overheads.

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## Enabling VLAN Support Using CLI Commands

You can use the `set network` command to set the primary IPv4 and IPv6 LAN1 networks as a tagged or an untagged VLAN interface.
Example:

```
Infoblox > set network
NOTICE: All HA configuration is performed from the GUI. This interface is
        used only to configure a standalone node or to join a Grid.
Enter IP address: 10.35.1.120
Enter netmask [Default: 255.255.0.0]: 255.255.0.0
Enter gateway address [Default: 10.35.0.1]: 10.35.0.1
Enter VLAN tag [Default: Untagged]: 110
Configure IPv6 network settings:
Enter IPv6 address [Default: 2620:10a:6000:2400::178]:
2620:010A:6000:2400:0000:0000:0000:6508
Enter IPv6 Prefix Length [Default: 64]: 64
Enter IPv6 gateway [Default: 2620:10a:6000:2400::1]:
2620:010A:6000:2400:0000:0000:0000:0001
Enter VLAN tag [Default: Untagged]: 110
NOTE: Configure of IPv4/IPv6 pure mode can be performed only via GUI.
Become grid member? (y or n):
```

You can execute the `show network` command to view the VLAN ID and tagged networks.
Example:

```
Infoblox > show network
Current LAN1 Network Settings:

IPv4 Address:               10.35.1.154
Network Mask:               255.255.0.0
Gateway Address:            10.35.0.1
VLAN Tag:                   Untagged
IPv6 Address:               2620:10a:6000:2400::19a/64
IPv6 Gateway Address:       2620:10a:6000:2400::1
IPv6 VLAN Tag:              Untagged
HA enabled:                 false
Grid Status:                Master of Infoblox Grid

Current Management Network Settings:

Management Port enabled:        true
Management IPv4 Address:        10.36.1.154
Management Netmask:             255.255.0.0
Management Gateway Address:     10.36.0.1
```

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```
Management IPv6 Address:        2620:10a:6000:2500::19a/64
Management IPv6 Gateway Address:2620:10a:6000:2500::1
Restrict Support and remote console access to MGMT port:   false

Current LAN2 Network Settings:

LAN2 Port enabled:                      true
NIC failover for LAN1 and LAN2 enabled: false
LAN2 IPv4 Address:                      10.34.71.188
LAN2 Netmask:                           255.255.255.0
LAN2 Gateway:                           10.34.71.1
LAN2 VLAN Tag:                          Untagged
LAN2 IPv6 Address:                      2620:10a:6000:22a3::bc/64
LAN2 IPv6 Gateway:                      2620:10a:6000:22a3::1
LAN2 IPv6 VLAN Tag:                     Untagged
```

To view VLANs for each port, you can use the `show interface` command. It displays the VLAN ID, tagged networks, and additional IP addresses that are configured for the network.

```
Infoblox > show interface
LAN1:

IP Address:  10.35.1.154       MAC Address: F4:87:71:00:07:05
Mask:        255.255.0.0       Broadcast: 10.35.255.255
MTU:         1500              Metric: 1
IPv6 Address:        2620:10a:6000:2400::19a/64
IPv6 Link:           fe80::f687:71ff:fe00:705/64
IPv6 Status:         Enabled
Negotiation: unknown
Speed:       1000M             Duplex: Full
Status:      UP BROADCAST RUNNING MULTICAST
SFP Type:    Fiber SX
SFP Model:   Finisar(FTLF8519P2BCL)

Statistics   Information

 Received

packets:  4715756 bytes:   287346518 (274.0 MiB)
errors:   0       dropped: 549
overruns: 0       frame:   0

Transmitted

packets:  21677   bytes:   15193301 (14.4 MiB)
errors:   0       dropped: 0
overruns: 0       carrier: 0
```

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```
Collisions:  0       Txqueuelen: 1000
```

It displays the VLAN details as follows:

```
LAN1 (VLAN Tag: 273):

IP Address: 10.34.80.188 MAC Address: F4:87:71:00:07:05
Mask: 255.255.255.0 Broadcast: 10.34.80.255
MTU: 1500 Metric: 1
IPv6 Address: 2620:10a:6000:22ac::bc/64
IPv6 Link: fe80::f687:71ff:fe00:705/64
IPv6 Status: Enabled
Negotiation: unknown
Speed: 1000M Duplex: Full
Status: UP BROADCAST RUNNING MULTICAST

Statistics Information

  Received

packets:  2964       bytes: 159373 (155.6 KiB)
errors:   0          dropped: 0
overruns: 0          frame: 0

  Transmitted

packets:  2272       bytes:   111809 (109.1 KiB)
errors:   0          dropped: 0
overruns: 0          carrier: 0

Collisions: 0    Txqueuelen: 0

Enter <return> for next page or q<return> to cancel the command.
```

You can use the `set default_route` command to specify an optional VLAN address and make it the default route. For more information, see the NIOS Documentation at *docs.infoblox.com,* CLI section.

# DSCP support

The DNS Cache Acceleration appliances support the configuration of DSCP (Differentiated Services Code Point) value for LAN1, LAN2, HA, MGMT, and VLAN interfaces. When the appliance is a member of an Infoblox Grid, the DSCP value for all interfaces is inherited from the Grid and you can override the DSCP value at the member level and at interface level. When you configure the DSCP value at the Grid or member level, all outgoing IP traffic on all interfaces uses the same value.

You can set the DSCP value for the primary LAN1 interface using the `set network` CLI command. For more information see the NIOS Documentation at *docs.infoblox.com,* CLI section. The DSCP values for all other interfaces and VLANs must be set through Grid Manager. For more information, see the NIOS Documentation at *docs.infoblox.com.* You can use the ping (-Q tos) and traceroute (-t tos) expert mode CLI commands, to prioritize all outgoing packets based on the DSCP value.

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## Configuring DSCP Value using CLI Commands

You can use the `set network` CLI command to set the DSCP value for the primary LAN1 interface.

> **Note**
> Once you configure the DSCP and enable the DNS acceleration on virtual platforms, the DSCP functionality is ignored for queries cached in DNS Cache Acceleration.

Example:

```
Infoblox > set network
NOTICE: All HA configuration is performed from the GUI. This interface is used only to
configure a standalone node or to join a grid.
Enter IP address:10.35.1.104
Enter netmask [Default: 255.255.255.0]:255.255.255.0
Enter gateway address [Default: 10.35.0.1]:10.35.0.1
Enter VLAN tag [Default: Untagged]:110
Enter DSCP value [Default: Inherited from Grid: 0]:25
Configure IPv6 network settings? (y or n):y
Enter IPv6 address [Default: none]: 2620:10A:6000:2400::168
Enter IPv6 Prefix Length [Default: none]: 64
Enter IPv6 gateway [Default: none]: 2620:10A:6000:2400::1
Enter VLAN tag [Default: Untagged]:110
Enter DSCP value [Default: Inherited from Grid: 0]:25
Become grid member? (y or n):
```

You can execute the `show network` command to view the DSCP values set for the primary LAN 1 interface.
Example:

```
Infoblox > show network
Current LAN1 Network Settings:
IP Address: 10.35.0.4
Network Mask: 255.255.0.0
Gateway Address: 10.35.0.1
VLAN Tag: 110
DSCP Value: 40
IPv6 Address: 2620:10A:6000:2400::4/64
IPv6 Gateway Address: 2620:10A:6000:2400::1
IPv6 VLAN Tag: 110
IPv6 DSCP Value: 40
HA enabled: false
Grid Status: Master of Infoblox Grid
```

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```
Current Management Network Settings:
 Management Port enabled:          true
 Management IPv4 Address:          10.35.15.104
 Management Netmask:               255.255.255.0
 Management Gateway Address:       10.35.0.1
 Restrict Support and remote console access to MGMT port:      false
```

**Note**
Additional addresses configured can be viewed using the `show interface` command.

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# Monitoring Cache Acceleration Statistics

The DNS Cache Acceleration appliances provide tools for monitoring cache acceleration statistics. You can use the following features to monitor and manage your appliance:

- *DNS Statistics for Cache Acceleration*
- *Supported Reports for DNS Cache Acceleration Appliances*
- *DNS Cache Acceleration CLI Commands*

## DNS Statistics for Cache Acceleration

The *DNS Statistics* widget on the Dashboard provides member-specific statistics or zone-specific statistics. The widget displays the totals for each type of DNS response as well as a line graph that tracks the responses per second.
You can add a *DNS Statistics* widget to your Dashboard for the appliance acting as a Grid member. To configure the *DNS Statistics* widget, click the Configure icon and do the following:

- Click **Select Member**. In the *Member Selector* dialog box, choose the DNS Cache Acceleration Grid member to display statistics for all its zones.

The widget displays only the option that you selected on your subsequent logins. When you click **Select Member**, the widget displays the **Select Member** option only, when you log in again.

- **Graph Configuration**: Select which DNS messages you want to track in the **Responses per Second** graph.
  - **Success**: The number of successful queries.
  - **NXDOMAIN**: The number of queries for domain names that did not exist in the database.
  - **Referral**: The number of queries that became referrals.
  - **NXRRSET**: The number of queries for domain names that did not have the requested records.
  - **Failure:** The number of queries that failed due to reasons other than nonexistent domain names or records in a domain.
  - **Recursion**: The number of recursive queries for which the name server sent queries to other name servers.

The widget displays the following information on three tabs:

- **DNS Responses** tab: Displays a pie chart and the total number of each type of message. If you enable DNS Cache Acceleration service on an appliance and if the appliance is active, then the data throughput from this service is displayed in this tab.
- **Responses per Second** tab: Displays a line graph that tracks the DNS responses received per second, within an hour. The displayed time is based on the time zone specified in the User Profile. If you enable the auto-detect time zone option in NIOS and Grid Manager cannot determine the browser time zone, the time is displayed in UTC format. You can mouse over the graph to display the coordinates of any point in the graph. If you enable the DNS Cache Acceleration service on Grid member that supports it, the data throughput from DNS Cache Acceleration service is displayed on this tab.
- **Cache Hit Ratio** tab: Displays a bar chart showing the cumulative cache hit ratio performance for the Grid DNS member. The widget measures BIND traffic cache hits, and combines with results from caching when the Grid member is a DNS Cache Acceleration appliance, or another caching server. The cache hit ratio (measured in percentage) is the total number of cache hits divided by the total number of cache lookups. The higher the ratio, the more effective the cache is at improving query performance.

# Reports for DNS Cache Acceleration Appliances

The DNS Cache Acceleration appliances display a list of predefined reports that you have selected when you set up your Grid or member reporting properties. You can modify user-defined reports, but you cannot modify or delete predefined reports or the search criteria of predefined reports.
To view all available reports, from the **Reporting** tab, select the **Reports** tab. The supported reports for DNS Cache Acceleration appliances are listed in the *table*. For information about how to create and manage user-defined reports, see the NIOS Documentation at *docs.infoblox.com*.

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**Note**
All DNS reports are supported on the DNS Cache Acceleration appliances. Some of these reports also include data collected from the caching hardware.

# Supported Reports for DNS Cache Acceleration Appliances

| DNS Reports | Security (DNS) Reports | System Reports |
|---|---|---|
| • DNS Response Latency Trend<br>• DNS Replies Trend<br>• DNS Cache Hit Rate Trend<br>• DNS Query Rate by Query Type<br>• DNS Query Rate by Server/ Server Group<br>• DNS Daily Peak Hour Query Rate by Server/Server Group<br>• DNS Query Trend Per IP Block Group<br>• DNS Top SERVFAIL Errors Sent<br>• DNS Top SERVFAIL Errors Received<br>• DNS Top Timed-Out Recursive Queries<br>• DNS Top Requested Domain Names<br>• DNS Top Clients<br>• DNS Top Clients Per Domain<br>• DNS Top NXDOMAIN/No ERROR (no data) | • DNS Top RPZ Hits<br>• DNS Top RPZ Hits by Client<br>• FireEye Alerts Report<br>• Threat Protection Event Count by Time<br>• Threat Protection Event Count by Severity Trend<br>• Threat Protection Event Count by Rule<br>• Threat Protection Event Count by Member<br>• Threat Protection Event Count by Member Trend<br>• Threat Protection Event Count by Category | • CPU Utilization Trend<br>• Memory Utilization Trend<br>• Traffic Rate Trend |

# DNS Cache Acceleration CLI Commands

Infoblox NIOS has commands that can be used to change settings related to DNS Cache Acceleration in an appliance that supports DNS Cache Acceleration. To use these commands, you must connect through SSH or connect through the serial console port to the NIOS appliance and log in with the appropriate admin account.
The NIOS appliance supports the standard NIOS command-line interface for communicating with the device. For more information see the CLI section in the NIOS Documentation at *docs.infoblox.com*.

The CLI commands listed below are frequently used on the DNS Cache Acceleration appliances:

- *set temp_license*
- *show license*
- *set dns*
- *show dns*
- *set monitor*
- *show monitor dns*
- *set dns-accel log*
- *set dns-accel log*
- *show dns-accel*
- *show dns-accel*
- *show dns-accel-cache*
- *show dns-accel-cache*
- *show dns-accel-cache [existing-file]*

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- *show dns-accel-cache [existing-file]*
- *set dns-accel-debug*
- *show dns-accel-debug*
- *set vdca_type64_support*
- *set vdca_type65_support*
- *show vdca_type64_support*
- *show vdca_type65_support*

## set temp_license

Generates and installs a temporary DNS cache acceleration license. You can add a temporary DNS Cache Acceleration license for up to one million QPS. Note that you cannot add a temporary license if a permanent license already exists. This command is used on software-based and hardware DNS Cache Acceleration appliances.

### Syntax

```
set temp_license
```

### Example

```
set temp_license
```

## show license

The `show license` command displays the capacity of the applicable DNS cache acceleration license type.

### Syntax

```
show license  [ all | csv | gridwide | revoked ]
```

### Example

`Infoblox > show license` all
`Infoblox > show license` csv
`Infoblox > show license` gridwide
`Infoblox > show license` revoked

## set dns

The `set dns` command enables you to control the DNS cache. You can flush the cache of a DNS view or flush a particular entry from the cache.

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### Syntax

```
set dns flush all [dns_view]
set dns flush name [dns_view]
```

| Argument | Description |
|---|---|
| `all` | Flushes the cache file from the default view. |
| `dns_view` | Specifies a particular DNS view. |
| `name` | Flushes the specific entry from the cache. |

### Example

```
Infoblox > set dns flush name accounting.corp100.com
Infoblox > set dns flush all
```

## show dns

The `show dns` command displays DNS query statistics for all DNS views. It also displays the recursive cache for the specified DNS views.

### Syntax

```
show dns {stats | cache [wait_time ntime][dns_view...]}
```

| Argument | Description |
|---|---|
| `stats` | Displays DNS query statistics for all DNS views. If the DNS Cache Acceleration service is enabled, **show dns stats** will show the combined statistics for BIND and DNS Cache Acceleration service. |
| `ntime` | The maximum time (from 1 to 600 seconds) to wait for the cache file to be ready. |

### Example

```
Infoblox > show dns stats
success 6718810
referral 0
nxrrset 0
nxdomain 0
recursion 257
failure 0
```

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## set monitor

This command activates the DNS monitoring in NIOS. This command is applied on software-based DNS Cache Acceleration appliance.

### Syntax

```
set monitor dns
```

### Example

```
Infoblox > set monitor dns
```

## show monitor dns

This command displays DNS network monitoring data. This command is applied on software-based DNS Cache Acceleration appliance.

### Syntax

```
show monitor dns
```

### Example

```
Infoblox > show monitor dns
```

## set dns-accel log

Sets the logging level for the DNS Cache Acceleration service on a DNS Cache Acceleration capable appliance. The logging level is between 0 being the least verbose and 7 being the most verbose.

### Syntax

```
set dns-accel log {level]
```

| Argument | Description |
|---|---|
| `level` | The logging level (from 0 to 7) of messages. |

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### Example

```
Infoblox > set dns-accel log 7
```

## set dns-accel log

Sets the logging level for the DNS Cache Acceleration service on software-based DNS Cache Acceleration appliance, where the level is between 0 being the least verbose and 7 being the most verbose.

### Example

Infoblox > `set dns-accel log 7`

## show dns-accel

The `show dns-accel` command checks the status and settings of the DNS Cache Acceleration service.

Note

When you use this command on a specific appliance, the configuration applies only to that appliance. When using this command for an HA pair, you must run the command separately on both the active and passive nodes to ensure consistent behavior.

The `show dns-accel` command displays the following operating characteristics of the DNS Cache Acceleration appliance.

| | |
|---|---|
| • Firmware version | • DNS query statistics |
| • Cache status, enabled/disabled | • DNS per-port packet count |
| • Cache hit count | • Log level |
| • Cache miss count | • Failed cores (if any) |
| • DNS minimum cached TTL | |

### Syntax

```
show dns-accel
```

### Example

```
Infoblox > show dns-accel
Firmware version:     1.3.19, Nov 08, 2011
```

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```
Cache:                Enabled
Cache hit count:      109397
Cache miss count:     3008
Minimum cached TTL:   10
DNS query stats:      SUCCESS=109380 NXDOMAIN=0 NXRRSET=0 FAILURE=0 REFERRAL=0
DNS UDP packet count: port0=299339 port1=93408 port2=0 port3=0
Log level:            6
Failed cores:         None
```

## show dns-accel

The `show dns-accel` command displays the following operating characteristics of the software-based DNS Cache Acceleration appliance:

```
Infoblox > show dns-accel
Log level:                      6(Info)
Cache:                          Enabled
Minimum cached TTL:             1
Maximum cached lifetime:        86400
Cache hit count:                0
Cache miss count:               0
DNS query stats:                SUCCESS=0 NXDOMAIN=0 NXRRSET=0 FAILURE=0 REFERRAL=0
System UDP DNS query count:     MGMT=0 LAN1=0 HA=0 LAN2=0
System UDP DNS response count:  0
System DOH/DOT query count:     MGMT=0 LAN1=0 HA=0 LAN2=0 BOND0=0
System DOH/DOT response count:  0
Infoblox >
```

You can view information about the SFPs on the DNS Cache Acceleration appliance. To view the details:

1. From the **Grid** tab, select the **Grid Manager** tab
2. Click the **Members** tab -> *member* check box -> *Detailed Status* icon.
   You can also use this command to view the VLAN ID, tagged networks, and additional IP addresses that are configured for the network when you enable VLAN on an DNS Cache Acceleration appliance. For more information about enabling VLAN support using CLI commands, see *Infoblox DNS Firewall*.

## show dns-accel-cache

You can view the current accelerator cache contents or view the prior cache data file. This command may affect DNS query response performance for around 15 seconds during execution. This command applies only to appliances with DNS cache acceleration license enabled.

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## Syntax

```
show dns-accel-cache
```

## Example

```
Infoblox > show dns-accel-cache
```

The following warning message is displayed, after you run this command.

WARNING

This operation will temporarily disable the acceleration cache to obtain the latest cached data.
The operation should take about 15 seconds to complete.
During this period, this member's DNS query performance may be significantly reduced, and changes to the member's DNS configuration will not be permitted.

1,*88ff8189f2f7548d.pacebook.com*,AAAA,IN,NO,0,1,3600,3557,6828,0,AA,AAAA,*88ff8189f2f7548d.pacebook.com*,3600,356f:3f49:366f:3f49:376f:3f49:386f:3f49,,0,2320,3557
2,*abe4d043815df77d.akamaidn.edu*,A,IN,NO,0,1,3600,3557,1151352,0,AA,A,*abe4d043815df77d.akamaidn.edu*,3600,52.111.63.73,,0,2321,3557
3,*7443b4a129f534b6.applexxx.net*,AAAA,IN,NO,0,1,3600,3557,1438037,0,AA,AAAA,*7443b4a129f534b6.applexxx.net*,3600,306f:3f49:316f:3f49:326f:3f49:336f:3f49,,0,2322,3557
4,*c42e7ae4ad44b227.pacebook.com*,AAAA,IN,NO,0,1,3600,3557,492849,0,AA,AAAA,*c42e7ae4ad44b227.pacebook.com*,3600,396f:3f49:3a6f:3f49:3b6f:3f49:3c6f:3f49,,0,2323,3557
5,*51bd345d7ac2a4b2.facebook.com*,AAAA,IN,NO,0,1,3600,3557,412,0,AA,AAAA,*51bd345d7ac2a4b2.facebook.com*,3600,3d6f:3f49:3e6f:3f49:3f6f:3f49:406f:3f49,,0,2324,3557

# show dns-accel-cache

The `show dns-accel-cache` command displays the operating characteristics of the software-based DNS Cache Acceleration appliance.

## Syntax

show dns-accel-cache

## Example

```
Infoblox > show dns-accel-cache

This operation should take about 15 seconds to complete. During this period, there
may be a slight impact on the DNS query performance for this member.
WRED feature also will be hit with this command.
Are you sure you want to proceed with this operation (y/n): y
```

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```
Cache data obtained at 2022-10-03-19:47:12
Cache is empty
```

## show dns-accel-cache [existing-file]

The `show dns-accel-cache existing file` displays the DNS Cache Information, fetches new acceleration cache data and displays it when using existing-file option.

### Syntax

```
show dns-accel-cache existing file
```

### Example

```
Infoblox > show dns-accel-cache existing file
```

1,*88ff8189f2f7548d.pacebook.com*,AAAA,IN,NO,0,1,3600,3557,6828,0,AA,AAAA,*88ff8189f2f7548d.pacebook.com*,3600,356f:3f49:366f:3f49:376f:3f49:386f:3f49,,0,2320,3557
2,*abe4d043815df77d.akamaidn.edu*,A,IN,NO,0,1,3600,3557,1151352,0,AA,A,*abe4d043815df77d.akamaidn.edu*,3600,52.111.63.73,,0,2321,3557
3,*7443b4a129f534b6.applexxx.net*,AAAA,IN,NO,0,1,3600,3557,1438037,0,AA,AAAA,*7443b4a129f534b6.applexxx.net*,3600,306f:3f49:316f:3f49:326f:3f49:336f:3f49,,0,2322,3557
4,*c42e7ae4ad44b227.pacebook.com*,AAAA,IN,NO,0,1,3600,3557,492849,0,AA,AAAA,*c42e7ae4ad44b227.pacebook.com*,3600,396f:3f49:3a6f:3f49:3b6f:3f49:3c6f:3f49,,0,2323,3557
5,*51bd345d7ac2a4b2.facebook.com*,AAAA,IN,NO,0,1,3600,3557,412,0,AA,AAAA,*51bd345d7ac2a4b2.facebook.com*,3600,3d6f:3f49:3e6f:3f49:3f6f:3f49:406f:3f49,,0,2324,3557

## show dns-accel-cache [existing-file]

The `show dns-accel-cache existing-file d` isplays the existing cache data file of software-based DNS Cache acceleration appliance.

### Syntax

show dns-accel-cache existing-file

### Example

```
Infoblox > show dns-accel-cache existing-file
Cache data obtained at 2022-10-03-19:47:12
Cache is empty
Infoblox > show date
Mon Oct  3 19:50:25 UTC 2022
Infoblox > show dns-accel-cache existing-file
Cache data obtained at 2022-10-03-19:47:12
Cache is empty
```

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```
Infoblox > show dns-accel-cache existing-file
Cache data obtained at 2022-10-03-19:47:12
Cache is empty
Infoblox > show date
Mon Oct  3 19:56:07 UTC 2022
```

## set dns-accel-debug

This command sets the maximum cached lifetime for DNS Cache Acceleration of software-based DNS Cache Acceleration appliance in seconds to a value range between 60 and 86400.

### Syntax

`set dns-accel-debug` <max-cached-lifetime> <value>

| Argument | Description |
|---|---|
| max-cached-lifetime | The maximum cached lifetime value. |

### Example

`Infoblox > set dns-accel-debug` **max-ttl 86400**

## show dns-accel-debug

This command displays the DNS Cache Acceleration configuration of software-based DNS Cache acceleration appliance.

### Syntax

`show dns-accel-debug [caching-conf | reporting-conf] [ tail [ <num> ] | /regex/ ]`

### Example

`Infoblox > show dns-accel-debug caching-conf tail 9 view_count8`

## set vdca_type64_support

This command allows you to enable or disable caching of DNS type 64 records. When the caching is enabled, the responses to DNS type 64 queries are cached in and returned from the virtual DNS Cache Acceleration cache.

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### Syntax

```
set vdca_type64_support [ on | off ]
```

| Argument | Description |
|---|---|
| on | Enables caching of DNS type 64 records. |
| off | Disables caching of DNS type 64 records. |

### Examples

Enabling caching of DNS type 64 responses:

```
Infoblox > set vdca_type64_support on
Enabled caching for Type 64 queries
```

Disabling caching for DNS type 64 responses:

```
Infoblox > set vdca_type64_support off
Disabled caching for Type 64 queries
```

## set vdca_type65_support

This command allows you to enable or disable caching of DNS type 65 records. When the caching is enabled, the responses to DNS type 65 queries are cached in and returned from the virtual DNS Cache Acceleration cache.

### Syntax

```
set vdca_type65_support [ on | off ]
```

| Argument | Description |
|---|---|
| on | Enables caching of DNS type 65 response. |
| off | Disables caching of DNS type 65 response. |

### Examples

Enabling caching of DNS type 65 responses

```
Infoblox > set vdca_type65_support on
Enabled caching for Type 65 queries
```

Disabling caching of DNS type 65 response

```
Infoblox > set vdca_type65_support off
Disabled caching for Type 65 queries
```

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## show vdca_type64_support

This command allows you to view whether the DNS type 64 responses are configured to be cached at virtual DNS Cache Acceleration. For example, `Enabled` returned in the output indicates that caching is enabled.

### Syntax

```
show vdca_type64_support
```

### Example

```
Infoblox > show vdca_type64_support
Type 64 support:         Enabled
```

```
Infoblox > show vdca_type64_support
Type 64 support:         Disabled
```

## show vdca_type65_support

This command allows you to view whether the DNS type 65 responses are configured to be cached at virtual DNS Cache Acceleration. For example, `Enabled` returned in the output indicates that caching is enabled.

### Syntax

```
show vdca_type65_support
```

This command has no arguments.

### Example

```
Infoblox > show vdca_type65_support
Type 65 support:         Enabled
```

```
Infoblox > show vdca_type65_support
Type 65 support:        Disabled
```

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